Rahul Goraniya

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Education

University of Birmingham *Birmingham, England*
B.Sc. in Computer Science *2022 – 2025*
Graduated with First Class Honours (GPA: 4.25/4.25)
Thesis ranked in the top 5% of the cohort

Research Projects

B.Sc. Thesis at the University of Birmingham *September 2024 – May 2025*
Supervisor: Dr. Rajesh Chitnis
Thesis: ETH-based Hardness for Approximation via L-Reductions from Matrix Tiling with $\leq$

- Proved an ETH-based PTAS lower bound of the form $2^{O\left((1/\varepsilon)^{\gamma}\right)} \cdot n^{O\left((1/\varepsilon)^{1-\delta}\right)}$ for Matrix Tiling with $\leq$.
- Used this framework to simplify existing PTAS lower bounds for Maximum Independent Set of Unit Disks.

Lower Bounds for Covering Problems *July 2024 – Present*
Collaborator: Dr. Rajesh Chitnis

- Established a tight ETH-based PTAS lower bound of the form $2^{O\left((1/\varepsilon)^{\gamma}\right)} \cdot n^{O\left((1/\varepsilon)^{1-\delta}\right)}$ for the *continuous* Covering Points with Squares problem.
- Prepared a full draft manuscript, currently being polished for submission to ICALP.

Parameterized Algorithms + Hardness *January 2025 – Present*
Collaborators: Dr. Rajesh Chitnis, Samuel Thomas, Kevin Kurien Alex

- Designed XP algorithms running in time $n^{O(k)}$ for the k-Disjoint $s \rightsquigarrow t$ Paths Problem with Min–Max and Balanced objectives, for both edge- and vertex-disjoint variants on DAGs.
- Proved matching lower bounds: the problem is W[1]-hard for both objectives and, assuming ETH, admits no algorithms with running time $f(k) \cdot n^{o(k)}$; the lower bound holds for planar DAGs in the edge-disjoint case and for 1-planar DAGs in the vertex-disjoint case as well.
- Manuscript in preparation, targeted for submission to ICALP.

Self-Reductions *September 2025 – Present*
Collaborators: Dr. Rajesh Chitnis, Dr. Namrata

- Studying self-reduction frameworks that yield hardness of approximation results without an explicit gap, using combinatorial and probabilistic constructions.
- Investigating which graph-theoretic problems admit such self-reduction based hardness proofs.

Relevant Coursework

Theoretical CS: Data Structures and Algorithms, Mathematics and Logic, Theories of Computation, Functional Programming in Haskell, Security and Networks (Cryptography), Algorithms and Complexity, Programming Languages: Principles, Design and Implementation, Advanced Functional Programming (Agda and Type Theory), Game Theory

Skills

Programming & Query Languages:	Agda, Haskell, Java, SQL, Neo4j, C
Tools and Technologies:	LaTeX, Git

Achievements

2025: Undergraduate thesis ranked in Top 5% of the cohort at the University of Birmingham

2025: Ranked 57th out of 399 students overall in the B.Sc. Computer Science cohort (Top ∼15%)

2022: CS International Excellence Scholarship: Awarded a £3,000 scholarship at the University of Birmingham

Extracurricular activities

- Studied the *Parameterized algorithms* textbook independently, gaining a strong foundation in core topics such as kernelization, algorithm design, graph theory, and lower-bound proofs.
- Conducted introductory self-study in Homotopy Type Theory and Category Theory, which deepened my knowledge and interest in foundations of mathematics and logic.
- Regularly attended the weekly theory seminars organized by the Mathematics and Theoretical Computer Science groups at the University of Birmingham. Notable talks included Dr. Sayan Bhattacharya's presentation on *Vizing's Theorem*, Dr. Saurav Chakraborty's work on *Distinct Elements in Streams*, and Dr. Parinya Chalermsook's overview of his research directions.
- Member of the Agda Theorem Proving Club during my final year, where we collaboratively formalized mathematical results from first principles using dependent type theory.